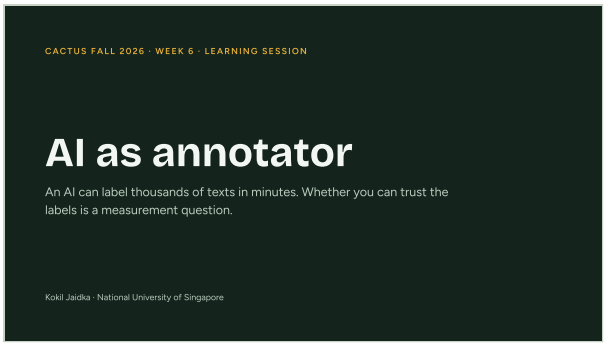
CACTUS Week 6: AI as Annotator



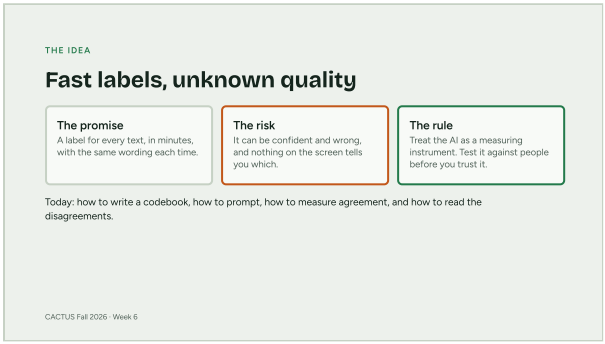
Slide 1

TIME 1 min (0:00)

SAY Welcome back. Last week you turned text into numbers with simple methods. Today an AI does the labeling, and your job is to find out whether you can trust it.

DO Welcome people. Post the course site link. Ask who has used an AI to classify or label text (hands or chat).

NOTE This is a 60-minute session. The measuring happens in the coding session.



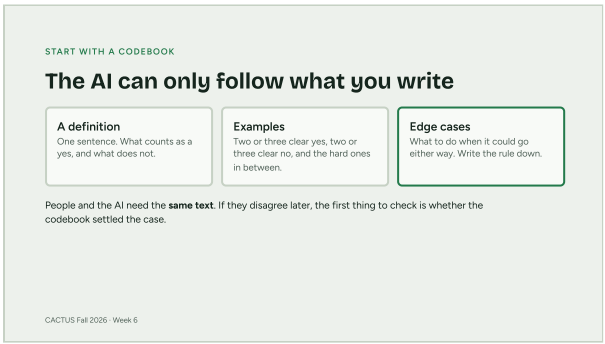
Slide 2

TIME 4 min (0:01 to 0:05)

SAY An AI can label thousands of texts in minutes, in the same words every time. That is the promise. The risk is that it can be confident and wrong, and nothing on the screen tells you which. So we treat it like a measuring instrument and test it against people.

DO Read the three cards. Say that today has four steps: codebook, prompt, measure, read the disagreements.

IF ASKED whether this replaces human coding say no. You still need people to define the codes and to check a sample. The AI scales the labeling, not the judgment.



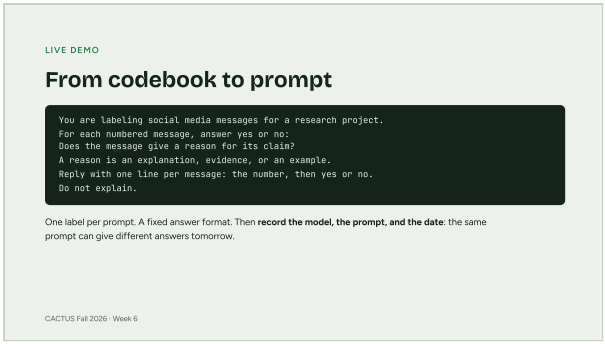
Slide 3

TIME 6 min (0:05 to 0:11)

SAY Before any prompt, write a codebook: a one-sentence definition, a few clear examples of yes and no, and a rule for hard cases. The AI can only follow what you write, and the people need the same text.

DO Ask the room for a definition of abusive, in one sentence. Take two. Point out how they differ: is criticizing a policy abusive? Is mocking a politician abusive? That is exactly what a codebook settles.

WATCH FOR People who skip this and go straight to the prompt. Say that disagreement later usually traces back to this slide.



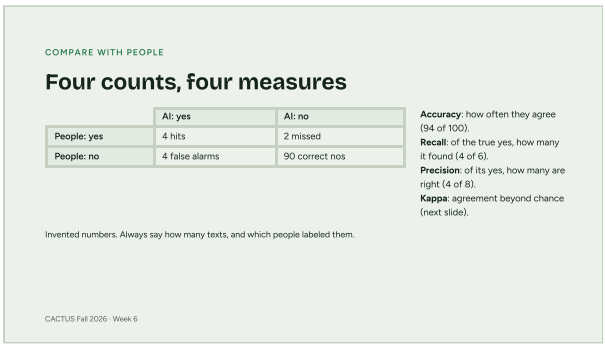
Slide 4

TIME 6 min (0:11 to 0:17)

DO (live demo) In an AI chat, paste the prompt on the slide plus six invented messages (prepare them before the session; do not use real posts or the class data). Show the answers. Then ask the room to label the same six by hand and compare.

SAY One label per prompt. A fixed answer format, so you can read the answers back in code. And record the model name, the prompt, and the date, because the same prompt can give different answers tomorrow.

WATCH FOR Do not paste the class data into the chat. Use invented messages. Do not reveal how the AI does on the real messages; students find that in the notebook.



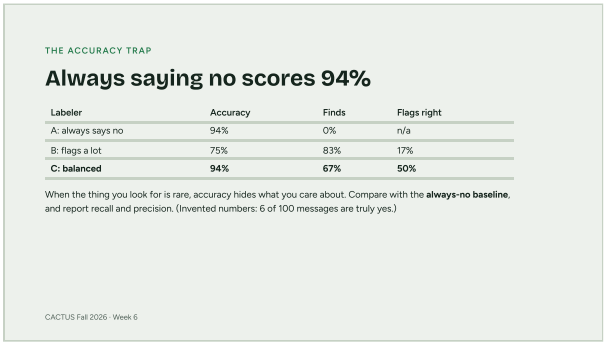
Slide 5

TIME 7 min (0:17 to 0:24)

SAY To compare the AI with people, count four things. Both say yes: a hit. People say yes and the AI says no: a miss. The AI says yes and people say no: a false alarm. Both say no: a correct no. From these come the measures. Accuracy is how often they agree. Recall is how many of the true yes it found. Precision is how many of its yeses are right.

DO Read the numbers on the slide. Say they are invented. Ask: which count would worry you more, misses or false alarms? It depends on what the labels are for.

IF ASKED for F1 say it combines precision and recall in one number; you can compute it, but read the two separately first.



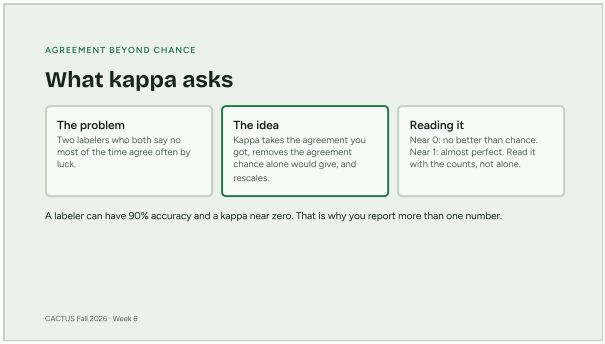
Slide 6

TIME 6 min (0:24 to 0:30)

SAY Here is the trap. Six of a hundred messages are truly yes. Labeler A says no every time: 94 percent accuracy, and it finds nothing. Labeler C also has 94 percent, and it finds four of six, with half of its flags right. Accuracy cannot tell them apart.

DO Ask the room which labeler they would use, and why. Say the practice page lets them try it. Say the rule: always compare with the always-no baseline, and report recall and precision.

WATCH FOR People who say B is the best because it finds the most. Ask them to look at its false alarms.



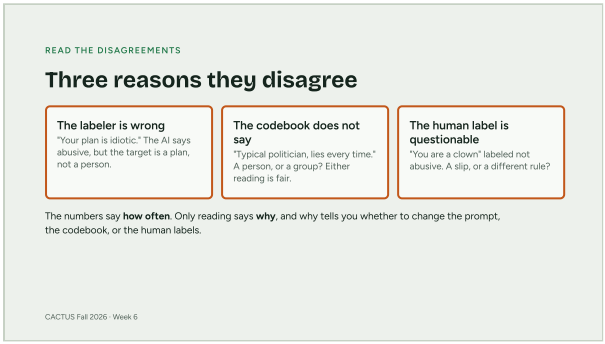
Slide 7

TIME 6 min (0:30 to 0:36)

SAY Two labelers who both say no most of the time agree often by luck. Kappa takes the agreement you got, removes the agreement that chance alone would give, and rescales. Near zero means no better than chance. Near one means almost perfect.

DO Say you can have 90 percent accuracy and a kappa near zero. The practice page has a slider so they can make exactly that. Say kappa is not a verdict: read it with the counts.

IF ASKED for a threshold say there are published rules of thumb, but they are conventions, and what is good enough depends on the use. Do not give a magic number.



Slide 8

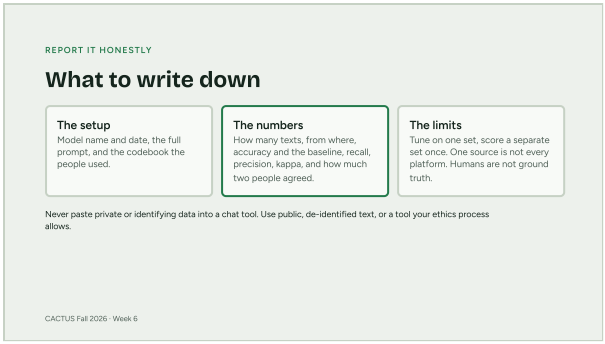
TIME 8 min (0:36 to 0:44)

SAY Numbers tell you how often the AI and the people disagree. Reading tells you why. There are three reasons. The labeler is wrong. The codebook does not say. Or the human label is questionable.

DO Read the three examples (all invented). Ask the room which one they would fix by changing the prompt, which by changing the codebook, and which by checking the human labels.

SAY (important) people are not ground truth. They are one group's judgments. Treat a disagreement as a question, not a verdict against the AI.

WATCH FOR Do not say what happens with the real messages in the notebook. Students find the surprises themselves.



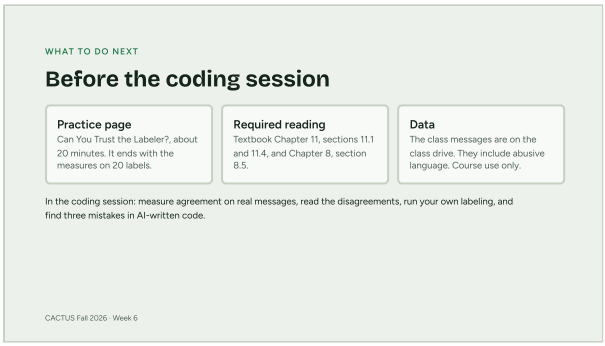
Slide 9

TIME 8 min (0:44 to 0:52)

SAY When you report, write down the setup: model name and date, the full prompt, the codebook the people used. Then the numbers: how many texts, from where, accuracy and the baseline, recall, precision, kappa. Then the limits: tune your prompt on one set of texts and score a separate set once, because a number from the texts you tuned on is optimistic. One source is not every platform. Humans are not ground truth, and the agreement between two people is a ceiling for the AI. Say what you did not test. Decide the agreement you would accept before you look at the result.

DO Say a sentence a good paper would write, for example: On 150 messages from one source, the AI agreed with people at kappa 0.3, with many false alarms; we did not test other sources. Ask the room to improve it.

SAY (important) do not paste private or identifying data into a chat tool. Use public, de-identified text, or a tool your ethics process allows.



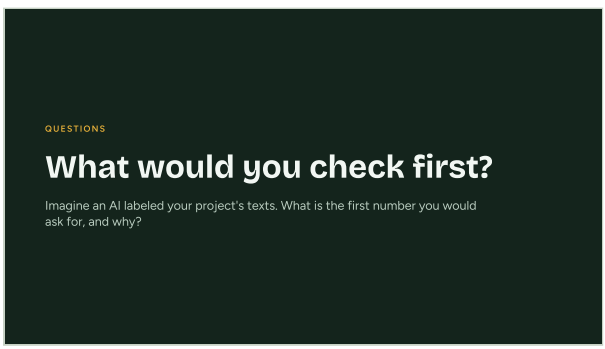
Slide 10

TIME 4 min (0:52 to 0:56)

SAY Before the coding session: the practice page (about 20 minutes), and the required reading (textbook Chapter 11, sections 11.1 and 11.4, and Chapter 8, section 8.5).

DO Post the practice page link and the course site link. Tell people where to get the class data (the class drive), that it includes abusive language, and that it is for course use only. Say that in one exercise they may paste up to 20 messages into an AI chat, and nothing else from the data.

SAY In the coding session there is no lecture. Measure agreement on real messages, read the disagreements, run your own labeling, and find three mistakes in AI-written code.



Slide 11

TIME 4 min (0:56 to 1:00)

SAY Imagine an AI labeled your project's texts. What is the first number you would ask for, and why? Write one sentence in your project card.

DO Take questions. If you do not know the answer, say so and write it down.

CLOSE Thank everyone. Say the next session is the coding session.